SparseSelfAttn: Efficient Linear-Complexity Transformers for NLP

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1. Abstract & Introduction

Modern Natural Language Processing relies on Transformer architectures utilizing multi-head self-attention mechanisms. In this study, we propose a sparse attention adaptation called SparseSelfAttn that reduces quadratic computational complexity O(N^2) to near-linear O(N sqrt(N)).

Our experiments on the GLUE benchmark achieve an overall score of 89.4%, outperforming standard BERT-Base while utilizing 38% less GPU memory during training.

2. Model Architecture & Theoretical Formulation

The core self-attention formulation computes attention weights across query, key, and value vectors: Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V. Standard models compute attention across all token pairs.

In SparseSelfAttn, attention connections are restricted to sliding local windows of size k=64 tokens combined with 16 global landmark tokens, preserving both contextual nuance and global document semantics.

3. Empirical Evaluation & Benchmarks

We evaluated our architecture on MNLI, QQP, SST-2, and SQuAD v2.0 datasets. On MNLI-matched accuracy, the model achieved 86.8%, representing a 1.2% improvement over baseline architectures.

Training was conducted on a cluster of 8 NVIDIA A100 GPUs (80GB VRAM) for 500,000 steps using AdamW optimizer with learning rate 3e-4, linear warmup for 10,000 steps, and cosine decay.

4. Limitations & Future Directions

While memory efficiency is substantially improved, our implementation currently lacks specialized kernel optimization for non-tensor core architectures.

Furthermore, extreme document lengths (>16,384 tokens) still exhibit slight perplexity degradation on long-form narrative coherence benchmarks.